Witnesses and orbits in finite incidence geometries

Private common-theory assembly; independent review pending

This draft assembles complete previously reviewed proof components into one common theory. A geometric shadow intersection must retain a flag witness; a subgroup intersection identity for distinguished generators is a different property. The elementary covering formulation, finite coset normalization, all-residue tests and sparse realization below make this distinction explicit. The common forest and thin-geometry arguments then support the two uniform applications reported in L096. No new theorem priority, whole-source completion or measurement is claimed by this unfinished assembly.

The source orbit and witness reductions are due to Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS Journal of Computation and Mathematics 7 (2004), 284–299, Section3. The block, stabilizer and transporter arguments are standard. The exact migration ledger distinguishes inherited proofs from the present reorganization; all paid common and retained classification premises remain explicit.

1 Flag-compatible covering

Let X be a finite set partitioned into type classes X_i, with a reflexive symmetric incidence relation $\sim$ such that distinct objects of the same type are not incident. A flag is a pairwise incident set, including the empty set. Define the common shadow of any set U by

$$\sigma_i(U) = \{y \in X_i : y \sim u \text{ for every } u \in U\}.$$

The query set need not itself be a flag. The geometric intersection test uses $U = \{x\} \cup F$, where F is a flag. It asks whether the nonempty set $S = \sigma_i(U)$ is the shadow of a flag incident with every object of U. A singleton S has its unique object as a witness; the whole set X_i has the empty flag as a witness.

Theorem 1. *Suppose $|S| \geq 2$, and put*

$$E = \{v \in X \setminus X_i : v \sim u \ (u \in U),\ v \sim s \ (s \in S)\}.$$

A witness flag exists if and only if some pairwise incident set $C \subseteq E$ meets every exclusion set $\{v \in E : v \not\sim y\}$, $y \in X_i \setminus S$.

Proof. A witness cannot contain an object of type i, since that would reduce its shadow to at most one object. Its incidence with U and its shadow S therefore place every member in E. Each y outside its shadow fails incidence with at least one member, giving the exclusion condition. Conversely, every eligible member is incident with S, so an eligible flag has shadow containing S. The exclusion condition removes every object outside S, proving equality. Eligibility also makes the flag incident with U. The argument includes $C = \varnothing$ when there are no objects to exclude. □

Equivalently introduce a Boolean variable z_v for each $v \in E$ and require

$$z_v + z_w \leq 1 \quad (v \neq w,\ v \not\sim w), \qquad \sum_{\substack{v \in E \\ v \not\sim y}} z_v \geq 1 \quad (y \in X_i \setminus S).$$

The first constraints include incompatible objects of the same type. Thus the criterion is an exact finite covering problem with flag compatibility retained. A solution certifies a witness; a heuristic failure is not a negative proof. The exhaustive finite algorithms below preserve the entire witness space.

Dropping compatibility is unsound, even for a rank-three geometry. Take $A = \{0, 1, 2, 3\}$, $B = \{b_0, b_1\}$ and $D = \{d_0, d_1\}$. Make every B object incident with every D object, and put

$$N_A(b_0) = N_A(d_0) = \{0, 1, 2\}, \qquad N_A(b_1) = N_A(d_1) = \{0, 1, 3\}.$$

Every vertex and cross-type edge extends to a chamber, so this is a geometry. For $x = b_0$, $F = \{b_1\}$, the target shadow is $S = \{0, 1\}$. The eligible set is $\{d_0, d_1\}$ and its common shadow is exactly S, but its two objects have the same type. No eligible flag has shadow S: the empty flag has four target objects and each singleton has three. The covering formulation and this obstruction reuse the project's previously reviewed proof; they are not a replacement claim to the original source's orbit algorithm.

2 Typed cosets and flag transitivity

The following construction is Tits' coset construction, recalled as Theorem 3.1 of Jacobs–Leemans. No finiteness of the group is needed here. Let I be a finite type set, G a group, and $(H_i)_{i\in I}$ subgroups. Take the *disjoint typed union* of the right-coset sets $H_i\backslash G$; the type remains part of an object even when two subgroups happen to be equal. Declare two typed cosets incident when their underlying subsets of G intersect.

Proposition 2. *This is a reflexive symmetric incidence structure, with no incident distinct objects of the same type. The cosets $(H_i)_{i\in I}$ form a chamber. Right multiplication by G preserves incidence and types and is transitive on flags of each prescribed type of size at most two. It need not be transitive on larger flags, nor need every flag extend to a chamber without an additional hypothesis.*

Proof. Right cosets of a fixed subgroup are either equal or disjoint. This gives the same-type assertion; reflexivity and symmetry follow from intersection. The displayed cosets all contain the identity and include one object of every type. Multiplication on the right is a bijection of G preserving intersections. Every object is a translate of its standard coset. If $H_i g$ and $H_j h$ are incident, choose a common element a; the two cosets equal $H_i a, H_j a$, respectively. Thus this pair is the translate by a of the standard pair. The empty flag is fixed. These arguments prove exactly the asserted transitivity. □

In the finite algorithms below, full flag transitivity is proved by checking all eligible-extension equalities, not inferred from this pair-transitivity statement. A nontrivial kernel of the coset action is allowed. Inversion identifies this right-coset convention with the left-coset convention used in the forest application, so no extra geometric premise is introduced when changing conventions.

3 Orbit and witness recognition

An incidence geometry has a finite set of types I, reflexive symmetric incidence, and no incident distinct objects of the same type. Every flag extends to a chamber. Suppose a finite group G acts transitively on flags of each type. Fix a chamber $C = \{p_i : i \in I\}$, use right actions, and write

$$H_i = G_{p_i}, \quad H_T = \bigcap_{t\in T} H_t, \quad H_\varnothing = G, \qquad C_T = \{p_t : t \in T\}.$$

The i-shadow $\mathrm{Sh}_i(F)$ is the set of type-i objects incident with every member of F. This geometric shadow property is distinct from the subgroup intersection property for distinguished generators of a C-group.

Lemma 3. *For every i, T, h we have*

$$\mathrm{Sh}_i(C_T^h) = (p_i^{H_T})^h.$$

Proof. If $i \in T$, both sides are the singleton $\{p_i^h\}$. Otherwise a type-i object y incident with C_T completes it to a flag of type $T \cup \{i\}$. Flag transitivity gives an element carrying $C_{T\cup\{i\}}$ to that flag while fixing each member of C_T. It belongs to H_T and carries p_i to y. Conversely every element of H_T fixes C_T, so its image of p_i is incident with C_T. Apply h to obtain the stated equality. □

The objects of type i are identified with right cosets $H_i\backslash G$. Thus the lemma uses one permutation action for each type. All shadows in that action are orbits of intersections of the chamber stabilizers. The recognition theorem in the current common theory for finite geometries can establish the required flag transitivity: for every $i \notin T$, the intersection $\bigcap_{t\in T} p_i^{H_t}$ must equal $p_i^{H_T}$. Induction on $|T|$ proves that this condition is sufficient, starting with the empty flag. It checks the entire set of eligible extensions, not a sample of chambers.

Theorem 4. *Fix a type i. The following condition is equivalent to the geometric shadow intersection condition at type i: for every element x and flag F, the intersection $\mathrm{Sh}_i(x) \cap \mathrm{Sh}_i(F)$ is empty or equals $\mathrm{Sh}_i(F')$ for a flag F' incident with both x and F.*

For each $j \neq i$, put $K = H_i \cap H_j$. For every nonempty $T \subseteq I \setminus \{i\}$, choose representatives h of

$$(H_i \cap H_T)\backslash H_i / K.$$

For each such representative form

$$X = p_i^{H_j} \cap (p_i^{H_T})^h.$$

If $|X| > 1$, there must be a subset $U \subseteq I \setminus \{i\}$ and an element $q \in K$ such that

$$(p_i^{H_U})^q = X, \qquad p_u^q \in (p_u^{H_T})^h \quad \text{for every } u \in U.$$

For a fixed U it suffices to take one representative of each right coset of $K \cap H_U$ in K.

Proof. Cases where x has type i, F contains type i, or F is empty have an immediate witness whenever the intersection is nonempty. Otherwise choose p in the nonempty intersection. The pair $\{p, x\}$ is a flag of types $\{i, j\}$; normalize it to $\{p_i, p_j\}$. The normalized F lies in the residue of p_i. Flag transitivity on $T \cup \{i\}$ expresses it as C_T^h with $h \in H_i$. Its stabilizer inside H_i is $H_i \cap H_T$. The remaining choices of normalization act by K, giving precisely the displayed double cosets. Lemma 3 identifies its intersection of shadows with X.

The singleton case has the witness consisting of its unique type-i object. If $|X| > 1$, a witness cannot contain type i. Let its type be U. Because $p_i \in X$, it is incident with p_i, and by assumption it is incident with p_j. Hence its union with $\{p_i, p_j\}$ is a flag. Flag transitivity expresses the witness as C_U^q for $q \in K$. Two elements of K give the same such flag exactly when they are in the same right coset of $K \cap H_U$. Lemma 3 gives the first displayed condition. Incidence with C_T^h says that, for every $u \in U$, its type-u object belongs to $\mathrm{Sh}_u(C_T^h)$, giving the second.

Conversely C_U^q is a flag. Since q fixes p_i and p_j, it is incident with those two objects. The second displayed condition makes it incident with C_T^h, and the first identifies its i-shadow with X. Transporting it back gives the required original witness. Every normalization and every orbit representative preserves these conditions, which proves sufficiency as well as necessity. □

There are two useful exact shortcuts. The candidate type U can be discarded unless $|p_i^{H_U}| = |X|$. Also, if p_j is incident with C_T^h and $|X|$ is the smaller of the two original shadow sizes, one of the original flags is already a witness. Both are consequences of the theorem and preserve its complete search space.

The source also omits certain candidate types in its implementation. These omissions have a direct justification. If a witness F' is incident with F and its type U is contained in the type T of F, the same-type incidence axiom forces every member of F' to be the corresponding member of F. Hence $F' \subseteq F$ and $\mathrm{Sh}_i(F') \supseteq \mathrm{Sh}_i(F)$. Such a type cannot realize an intersection strictly smaller than $\mathrm{Sh}_i(F)$. If the intersection has the smaller of the two original sizes but x is not incident with F, a candidate of type $U = T$ would have to equal F and so would fail incidence with x. It can also be omitted. These prove the two type-pruning steps in the source Figure 1. The present complete search may retain those impossible candidates; doing so changes time but not the answer.

4 Every residue and every rank

For $J \subseteq I$, let R_J be the residue of the base flag C_J. Its types are $I \setminus J$. Empty residues are not tested for an existential type condition.

Theorem 5. *Suppose the coset system is flag transitive. The map*

$$(H_J \cap H_i)h \longmapsto H_i h, \qquad h \in H_J, \quad i \in I \setminus J,$$

is an isomorphism of the coset geometry inside H_J with R_J. In particular H_J is flag transitive on R_J. For each $k \in \{2, \dots, |I|\}$ it suffices to test precisely the base residues R_J with $|I \setminus J| = k$. On each of them apply Theorem 4 with group H_J and parabolics $H_J \cap H_i$. All type tests in all these residues give $(\mathrm{IP})_k$*; at least one type test in each residue gives* $(\mathrm{WIP})_k$*. The conjunction over $k = 2, \dots, |I|$ gives respectively IP and WIP.*

Proof. An object of type i incident with C_J extends that flag. Flag transitivity therefore writes it as $H_i h$ for $h \in H_J$. Two such representatives give the same object exactly when they give the same right coset of $H_J \cap H_i$ inside H_J. This proves object bijectivity. If the small cosets intersect, their common representative gives incidence of their images. Conversely, two incident images together with C_J form a flag. Flag transitivity gives a representative in H_J for both images, so their small cosets intersect. The same argument, applied to an arbitrary flag of the residue together with C_J, proves flag transitivity under H_J.

Every flag of type J is a translate of C_J, so all its residues are isomorphic with their type labels retained. This proves that one base residue per subset is sufficient; it does not identify different subsets or assume their residues are isomorphic. Inside a fixed residue the preceding theorem is an equivalence at each remaining type. Universal quantification of those types is IP at that rank; the existential type for WIP is chosen separately in each residue. These are precisely [JL, Lemmas 2.1–2.2]. Rank-one shadow intersections are empty, singletons, or the whole sole type, with the corresponding immediate witnesses. Rank zero has no available type and is excluded, as in those lemmas. □

There are $[G : H_J]$ flags of type J, so the report records that many actual residues represented by each base residue. It computes every remaining type, even when one failed type would already refute IP. Thus its WIP output does not silently use a fixed global type.

5 Geometry hypotheses are separate checks

The supplementary rows are not accepted because their total matches a published table. They define coset incidence structures, which need not be flag transitive. We first check the orbit

equality preceding Theorem 4 for all i, T. This condition is necessary by Lemma 3. For sufficiency, normalize an existing flag of type T inductively. Every object eligible to extend it belongs to the intersection of the singleton shadows, hence to the H_T orbit by the checked equality. It follows that each extended flag is a translate of the corresponding base flag. Induction covers every flag type, and in particular every flag extends to a chamber.

Write $K = H_J$ and $P_i = K \cap H_i$ for a residue. The following tests are the actual source hypotheses, with kernels of actions allowed.

Lemma 6. *For a flag-transitive coset geometry the following conditions are exact.*

1. *Firmness holds if and only if $[H_{I\setminus\{i\}} : H_I] \geq 2$ for every i.*
2. *Residual connectedness holds if and only if $\langle P_i : i \in I \setminus J\rangle = K$ for every residue of rank at least two.*
3. *Residual weak primitivity holds if and only if in each residue of positive rank at least one transitive action on $P_i\backslash K$ is primitive of degree at least two.*
4. *The local condition $(2T)_1$ holds if and only if, in every rank-one residue, its point stabilizer has exactly two orbits on its objects and there are at least two objects.*

Proof. The objects completing the base panel missing type i are the cosets of H_I in $H_{I\setminus\{i\}}$. Flag transitivity covers every panel, proving the first claim. For the second, a coset path starting at the base chamber can multiply its representative successively by elements of the P_i. Conversely each such multiplication is realized by a path through the object fixed by that subgroup. Hence the connected component of the base chamber is exactly the orbit of their join, and all components are its cosets. The argument is applied inside each K; it is valid for rank two as well as higher ranks. The third statement is the definition of weak primitivity applied to each residue. Equivalently P_i must be maximal proper in K: blocks containing the base coset correspond to intermediate subgroups. This equivalence does not require a faithful action. Finally a transitive action is two-transitive exactly when the stabilizer of a point is transitive on the other points. Its two orbits are then that point and its complement. Again this depends on the permutation image, not on a faithful ambient action. □

These are the panel and coset-path arguments already used in the accepted Hall–Janko census [D]. The implementation tests the entire action, not only subgroup orders or a presumed diagram.

6 Rank two without a diagram assumption

In a rank-two incidence structure the only nonautomatic shadow test compares the neighborhoods of two distinct objects on the same side. At a fixed shadow type i, it passes exactly when every such intersection has size zero, one, or the whole type-i set. Indeed a witness of type i has singleton shadow; a witness of the other type cannot be incident with both distinct objects of that type. The empty flag has the whole shadow set. These are all possible witness flags. In particular comparing an object with itself is automatic and is excluded from this pair test.

Both type tests hold if and only if the graph has no four-cycle or is complete bipartite. If a four-cycle is present, its two vertices on one side have at least two common neighbors, hence all vertices of the other side. Every pair on that other side consequently has at least two common neighbors and must meet the entire first side. This proves completeness. The converse follows immediately. This is the accepted direct proof [Cover], recovering the partial-linear-space/generalized-digon criterion used in [JL, Section 3]. We still record the two directional tests separately, and do not replace the weak condition by an assumed diagram.

For a coset action, fix the first object of the opposite type j. Representatives of $P_j\backslash K/P_j$ cover its possible second objects. Discard the identity double coset, which represents the same object. Compute the two neighborhoods by their actual coset-action orbits. Their intersections then perform the criterion just proved.

7 A sparse realization of the common shadow theorem

For a finite permutation group K and subgroups $P_1, \dots, P_r$, write $P_T = \bigcap_{t\in T} P_t$ and $P_\varnothing = K$. The i-shadow of the base flag of type T is $\{P_i x : x \in P_T\}$. The original implementation realizes the whole right-coset action of degree $[K : P_i]$. The following representation avoids constructing a large universe when only small shadows are used.

Proposition 7. *For nonempty T, take a right transversal of $P_i \cap P_T$ in P_T and replace each representative x by a canonical representative of $P_i x$. This set represents the entire i-shadow without duplicates. Its right translate by g is obtained by replacing each representative x by the canonical representative of $P_i xg$. The empty flag may instead be represented by the cardinality $[K : P_i]$, provided its universal-set role is handled explicitly as below.*

Proof. For $x, y \in P_T$, equality $P_i x = P_i y$ holds exactly when $xy^{-1} \in P_i \cap P_T$. Thus the transversal gives exactly one representative per shadow object. Canonicalization changes only its name. If $x' = hx$ with $h \in P_i$, then $P_i x'g = P_i xg$, so the translated representative is well defined and realizes the same right action. Intersections, containment, and membership are consequently unchanged.

The empty flag has the full object set as shadow. Its flag-extension identity is tautological and is counted as such. In a witness test, the computed intersection is already a subset of the i-object set. It is therefore the empty flag's shadow exactly when its cardinality is $[K : P_i]$; the empty flag is automatically incident with both tested flags. When this equality fails, the empty flag cannot witness the intersection, and every nonempty witness is tested by the displayed representation. No intersection with a fictitious empty set is performed. The same degree is used in the rank-two complete-bipartite exception. □

The normalization, double-coset enumeration and flag-constrained witness search are exactly those of the accepted common theorem. Only the representation of each finite shadow changes. Trivial panel stabilizers are allowed: the subgroup generated by an empty generator list is formed with the identity included, so a two-object panel is still recognized. The original accepted implementation remains unchanged.

8 Primitive actions and exact family transport

8.1 Primitivity without constructing a large coset action

Let $H \le G$ be finite. The transitive action of G on its right cosets of H is primitive of degree at least two if and only if $H < G$ and

$$\langle H, x\rangle = G \quad \text{for every representative } x \notin H \text{ of an } (H,H)\text{-double coset in } G. \tag{*}$$

This is the standard block–overgroup correspondence together with an explicit double-coset test. Here is the complete argument. A block containing the base coset H is the orbit of H under a subgroup K with $H \le K \le G$: its setwise stabilizer is K, and transitivity shows the block is exactly $\{Hk : k \in K\}$. Conversely this set is a block for each such K. Hence degree-at-least-two primitivity is exactly maximal properness of H. If $g = h_1 x h_2$, with $h_1, h_2 \in H$, then $\langle H, g\rangle = \langle H, x\rangle$. A proper intermediate subgroup therefore contains an element outside H and supplies a representative violating (*); conversely any violating representative gives the proper

intermediate subgroup $\langle H, x\rangle$. The argument does not require a faithful coset action or a core-free stabilizer. The case $H = G$ must be rejected separately: its degree-one action is outside the convention and would otherwise satisfy (*) vacuously.

The finite implementation enumerates the complete double-coset roster, forms each indicated generated subgroup and compares its order with $|G|$. Every negative nontrivial case records the outside element and actual intermediate subgroup generators. For a residue it tests types in increasing coset degree until a primitive type is found, or exhausts all types. This preserves the existential weak-primitivity quantifier. Applying it to all base residues is legitimate by the separately checked flag transitivity; the residue common theorem supplies their actual stabilizers and translates. No previously classified geometry predicate is used by this eligibility test.

8.2 Exact family transport with necessary order filters

Let $A \leq \mathrm{Aut}(G)$ be a finite, explicitly computed transformation group, and let $\mathcal{H} = (H_1, \ldots, H_n)$ and $\mathcal{K} = (K_1, \ldots, K_n)$ be ordered families. We seek an element of A carrying the first family to a permutation of the second. Put $H_J = \bigcap_{j\in J} H_j$, with $H_\varnothing = G$, and define K_J similarly. If a proposed type permutation sends H_i to $K_{\pi(i)}$, then necessarily

$$|H_J| = |K_{\pi(J)}| \qquad (J \subseteq \{1, \ldots, n\}). \tag{**}$$

Indeed an automorphism commutes with intersection and preserves order. Thus a failed equality rejects that type permutation, and a failed comparison of the two multisets of intersection orders rejects every permutation. Satisfying (**) never certifies equivalence.

Fix a surviving target tuple $\mathcal{T}$ and any order $i_1, \ldots, i_n$ in which to place its entries. Start with $C = A$ and $\mathcal{H}' = \mathcal{H}$. At step k find an actual transporter $x \in C$ with $x(H'_{i_k}) = T_{i_k}$. If none exists, reject this target. Otherwise replace every H'_i by $x(H'_i)$ and replace C by the setwise stabilizer of T_{i_k} in C. After the last step no further stabilizer is needed. Success supplies a product of the computed automorphisms taking the entire family to the target.

For completeness, after the first $k-1$ placements all possible further solutions differ from the current partial transporter by an element of the pointwise stabilizer of the already placed *subgroups*. If x is one solution to the next placement, all such solutions differ from x by an element fixing its target subgroup. Hence passing to the stated stabilizer loses no solution. Induction proves both soundness and completeness, for every placement order. The stabilizer after k placements depends only on A and the first k target subgroups, not on the input family or the transporters chosen. It can therefore be cached for repeated comparisons against the same target and placement order. In this implementation larger target subgroups are placed first, with indices breaking ties. This changes computation time, not the criterion. Literal equality of complete tuples is an immediate successful case.

8.3 Reducing automorphisms to simultaneous conjugacy

There is a useful version when a complete set B of representatives for $\mathrm{Aut}(G)/\mathrm{Inn}(G)$ is available. For each $\beta \in B$ and each target type permutation, apply β to the whole input family. The remaining question is simultaneous conjugacy by one element of G. In the preceding procedure start with $C = G$; replace subgroup transport under automorphisms by conjugation, and replace each target stabilizer by its normalizer in C.

Every automorphism is an inner automorphism composed with one tested representative, since the inner automorphisms form a normal subgroup. For a fixed representative, the same transporter-coset argument proves that the successive normalizers preserve exactly all possible remaining conjugators. A positive result records the representative's images of the ambient generators, one actual conjugating element, and the type permutation; together these identify the entire family. Neither a splitting of the automorphism group nor a representative of the same order as its quotient image is required. In particular, when the inner subgroup has index

two, any verified noninner automorphism together with the identity is a sufficient complete transversal. This is a standard quotient and transporter reduction; its implementation below is not a new classification theorem.

9 Forest subgroups and their incidence geometry

Let T be a finite tree on a set Ω of $N \geq 2$ vertices, with edge set E. For $F \subseteq E$, let K_F be the subgroup of $\mathrm{Sym}(\Omega)$ generated by the endpoint transpositions of the edges in F. Set $H_e = K_{E\setminus\{e\}}$. The elements of type e in $\Gamma(T)$ are left cosets gH_e. Two such elements are incident when their cosets intersect; incidence is reflexive, and distinct elements of one type are not incident. A flag is a set of pairwise incident elements. The residue of a flag consists of its incident elements of the remaining types.

Lemma 8. *If C ranges over the components of (Ω, F), then*

$$K_F = \prod_C \mathrm{Sym}(C), \qquad K_F \cap K_D = K_{F\cap D}.$$

The coset system $\Gamma(T)$ is a thin, residually connected geometry on which $\mathrm{Sym}(\Omega)$ is flag-transitive. The residue at the standard flag of types $E \setminus F$ is the join of the geometries of the nontrivial components of (Ω, F).

Proof. Transpositions along a connected graph generate its full symmetric group: conjugating successive edge transpositions along a path produces the transposition of its endpoints. Distinct components have disjoint supports. A permutation in both K_F and K_D preserves every block of the common refinement of their component partitions. Two vertices are in such a block exactly when their unique tree path belongs to both forests, hence to $F \cap D$. This proves the two group formulas.

Root T, and let A_e be the component of $T - e$ away from the root. The coset gH_e identifies with the subset gA_e of Ω. For distinct types, incidence is precisely

$$|X_e \cap X_f| = |A_e \cap A_f|.$$

Indeed these intersection sizes and the two fixed subset sizes determine all four atoms of the pair; equal atom sizes give a permutation carrying one pair to the other. The A_e form a laminar family. Thus pairwise incidence forces exactly their nesting and disjointness relations on the X_e. Adjoin Ω above the selected sets and subtract each container's maximal selected proper subcontainers. The resulting atoms partition Ω and have the same sizes on the standard and target sides. Bijections of corresponding atoms combine to one permutation realizing the entire flag. Every flag therefore extends to a chamber, and flags of each fixed type form one orbit.

The stabilizer of the standard flag of types $E \setminus F$ is K_F. Its maximal parabolics for the remaining types are $K_{F\setminus\{e\}}$. Their direct product descriptions identify the residue with the claimed join; isolated vertices contribute trivial factors. The chamber stabilizer is $K_\varnothing = 1$, and each panel stabilizer has order two, so the geometry is thin. In any residue of rank at least two, the maximal parabolics collectively contain every remaining edge transposition, and hence generate the residue group. Their coset incidence graph is connected: products of subgroup elements give paths between standard and arbitrary cosets. This proves residual connectedness. □

For an element type i, write $\sigma_i(F)$ for the set of elements of type i incident with every member of F. The *intersection property* requires that for every element x and flag F, a nonempty $\sigma_i(x) \cap \sigma_i(F)$ be $\sigma_i(F')$ for a flag F' incident with x and F. This is required in the geometry and in every residue. It is distinct from the subgroup intersection identity of Lemma 8.

10 Explicit witnesses for Boolean shadows

Let $\mathcal{B}(\Omega)$ have the nonempty proper subsets of Ω as elements, cardinality as type, and comparability by inclusion as incidence.

Lemma 9. *Fix $0 < k < N$. For any family $\mathcal{U}$ of elements of $\mathcal{B}(\Omega)$, a nonempty common k-shadow has a witness flag of at most two elements, incident with every member of $\mathcal{U}$.*

Proof. If $\mathcal{U}$ contains a k-subset X, every common shadow element must equal X. Nonemptiness says that X is incident with the whole family, so $\{X\}$ is the required flag. Otherwise put

$$L = \bigcup_{\substack{X \in \mathcal{U} \\ |X| < k}} X, \qquad V = \bigcap_{\substack{X \in \mathcal{U} \\ |X| > k}} X,$$

with empty union $\varnothing$ and empty intersection Ω. The common shadow is exactly

$$\{Y \subseteq \Omega : |Y| = k,\ L \subseteq Y \subseteq V\}. \tag{1}$$

Nonemptiness gives $L \subseteq V$ and $|L| \leq k \leq |V|$. If either cardinality bound is equality, the shadow has a unique element, which again supplies a singleton flag incident with the whole family. In the remaining case use $\{L, V\}$, omitting $\varnothing$ and Ω. Its members are proper subsets of different sizes and form a flag. Every smaller query member is contained in L, and every larger one contains V, so both witness elements are comparable with every query member. Its shadow is precisely (1), including when one or both boundary sets were omitted. □

Lemma 10. *The geometry $\mathcal{B}(\Omega)$ satisfies the intersection property in all residues. Joins of geometries with that property also satisfy it.*

Proof. Apply Lemma 9 to $\mathcal{U} = \{x\} \cup F$ for the property in $\mathcal{B}(\Omega)$. A fixed chain, completed by the formal endpoints $\varnothing$ and Ω, divides its residue into the proper intervals between successive members. An interval (A, B) is the Boolean geometry on $B \setminus A$, via $X \mapsto X \setminus A$; the types are shifted by $|A|$. All elements in different intervals are incident, so the residue is their join. Empty intervals add no types. A rank-one factor has only empty, full or singleton shadows; the empty flag witnesses a full shadow, and its unique point witnesses a singleton.

For completeness, in a join fix a type i in one factor. Only query elements of that factor constrain its shadow. If x belongs to it, use the factor's intersection property with the part of F in that factor. If x belongs to another factor, the part of F in the target factor is itself a witness, including when it is empty. The witness is automatically incident with all query elements in other factors. Residues of a join are joins of factor residues, proving the assertion also at every residue. □

Theorem 11. *The geometry $\Gamma(T)$ satisfies the intersection property if and only if T is a path.*

Proof. For a path, root at one endpoint. Its cuts A_e form a chain with sizes $1, \ldots, N-1$. The subset description in Lemma 8 identifies its incidence with comparability, so $\Gamma(T) = \mathcal{B}(\Omega)$. Lemma 10 proves the positive assertion.

A tree which is not a path has three edges incident at a common vertex. Keep precisely those edges in Lemma 8. The resulting residue is the geometry of the claw, with three types a, b, c, each a copy of $\{1, 2, 3, 4\}$. Elements of different types are incident exactly when their labels differ. Take $x = (b, 1)$ and $F = \{(b, 2)\}$, with target type a. Their common shadow is $\{(a, 3), (a, 4)\}$. A witness cannot contain an a-element, which would give at most a singleton shadow, or a b-element, which cannot be incident with both distinct b-elements. It can contain at most one c-element. The empty flag has four shadow elements, whereas any c-element has three. Neither has the required shadow. Thus this residue fails the property. □

11 Thin rank-three geometries and involution triples

Throughout, the group preserves types and acts transitively on flags of each type. Incidence is symmetric and reflexive, with no incident distinct objects of one type. Every flag extends to a chamber. A geometry is thin if every flag missing one type has exactly two extensions. Residual connectedness requires the incidence graph of every residue of rank at least two to be connected. Residual weak primitivity means that in every nonempty residue at least one of its actions on a type is primitive.

Lemma 12. *Let a nonabelian finite simple group G act as above on a thin, residually connected geometry of rank three. For a chamber $C = \{p_0, p_1, p_2\}$, write $H_i = G_{p_i}$ and $B = \bigcap_i H_i$. There are distinct involutions t_0, t_1, t_2 such that*

$$B = 1, \qquad G = \langle t_0, t_1, t_2 \rangle, \qquad H_i = \langle t_j : j \neq i \rangle, \qquad H_i \cap H_j = \langle t_k \rangle \quad (\{i, j, k\} = \{0, 1, 2\}).$$

Proof. Put $R_i = \bigcap_{j \neq i} H_j$. Flag transitivity on the two extensions of $C \setminus \{p_i\}$ gives $[R_i : B] = 2$, so B is normal in each R_i.

The chamber graph is connected. To see this, a rank-two residue has a connected incidence graph, whose edges are its chambers; successive edges sharing a vertex give a gallery. In rank three take a path of incident objects between a vertex of one chamber and a vertex of another. Extend each edge to a chamber. Consecutive chosen chambers contain a common object, and the rank-two residue of that object supplies a gallery between them. The end chambers connect in the same way. Thus all chambers are connected by changes of one object.

The chambers adjacent to C across its i-panel are images of C under R_i. Transporting along a gallery proves $G = \langle R_0, R_1, R_2 \rangle$: the generated subgroup reaches every chamber, and it contains their common base stabilizer B. Therefore $B \lhd G$. Since $B < R_i$, simplicity gives $B = 1$. Each R_i now has a unique nonidentity element t_i, an involution. Within the residue of p_i, the same gallery argument gives $H_i = \langle R_j : j \neq i \rangle$. Distinctness follows because an element in two R_i lies in B. The stated intersections follow from the stabilizer definitions. □

Lemma 13. *Conversely take three distinct involutions generating G, and put $H_i = \langle t_j : j \neq i \rangle$. Suppose the three intersections in Lemma 12 hold. Their coset incidence system is a flag-transitive geometry if and only if, for each ordering i, j, k of the types,*

$$(H_i H_j) \cap (H_i H_k) = H_i (H_j \cap H_k). \tag{2}$$

It then is thin and residually connected. It is residually weakly primitive if and only if at least one H_i is maximal in G and all three numbers $|H_i|/2$ are prime.

Proof. Any incident pair of cosets has a common element, so G is transitive on incident pairs. Normalize the cosets of types j, k to H_j, H_k. The cosets $H_i g$ incident with both correspond exactly to the elements of $(H_i H_j) \cap (H_i H_k)$. They extend to chambers in the base chamber's orbit precisely when they meet $H_j \cap H_k$. This gives (2). Every flag then extends to such a chamber. The right side is contained in the left side; since the triple intersection is trivial, its size is $2|H_i|$. Thus equality is checked without constructing an entire incidence graph.

The base chamber stabilizer is trivial and its panel stabilizers have order two, so the geometry is thin. Each H_i is generated by the two panel stabilizers within it; the whole group is generated by all three. Their coset incidence graphs are connected, proving residual connectedness. Each rank-two residue is the polygon associated to a dihedral group of order $2m_i$, where $m_i = |H_i|/2$. Its vertex stabilizers are reflections. A reflection subgroup is maximal in a dihedral group of order $2m_i$ exactly when m_i is prime: intermediate subgroups correspond to proper divisors of the rotation order. Rank-one residues have degree two and are primitive. The remaining condition is primitivity on some type of the whole geometry, equivalent to maximality of its stabilizer H_i. □

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[Cover] The project's accepted phase-four covering proof and later four-lemma formal supplement. The latter proves only the flag-compatible covering equivalence, not the coset computations.

[D] The project's previously accepted common finite-geometry proofs and exact source-specific census packets. Each application names which census completeness and group models it retains.

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